Sign-Schur, reduced

Dirichlet zones of the Christoffel–Darboux kernel, not a bipartition of the ν-nodes

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August 28, 2026

Abstract

Round 392 located the 99.7% Assist cancellation in the *signs* of the Uvarov image $K^\mu[\Xi]$, not in the positive τ-quotients of occupied Fejér. The candidate mechanism was a source-defined checkerboard along the ordered ν-nodes, inherited from the oscillating Chebyshev–Christoffel–Darboux kernel, which via a diagonal sign conjugation $D_\sigma E D_\sigma$ would force $\lambda_{\max}(E_k) \le \mathrm{maxdiag}\cdot(1+o(1))$. What is proved: on the cosine grid the Chebyshev-T kernel is the Dirichlet combination of [2], the sign of the difference term $D_{n-1}(\theta_i-\theta_j)$ is zonal (depends only on which π-interval $(n-\frac{1}{2})|\Delta\theta|$ occupies), and $|D_n(\alpha)| \le \min(2n+1, 1/|\sin(\alpha/2)|)$. The hoped M-matrix bound is false even in 2×2: $\left(\begin{smallmatrix} 1 & -c \\ -c & 1 \end{smallmatrix}\right)$ has $\lambda = 1+c > \mathrm{maxdiag}$. What is *not* proved: rank-1 conjugability. Mass-weighted checkerboard agreement is a coin (0.504 on FRAME-A $w=9$; core-42 in $[0.429, 0.512]$). The large couplings *do* inherit Chebyshev signs (mass-weighted 0.810 on FRAME-A, $[0.810, 0.894]$ on core-42), but the 11–19% residual plus the failed bipartition block every source-defined $k^* > 2N/5$. Two-period lag-1 is in-phase (named kill). Scramble holds the mass-map and dies on the envelope. No RH claim.

Status. Lemmas 1–3 are proved (finite identities / a named counterexample). Lemma 4 is a machine identity on the two-period mesh. Propositions 1–2 are a finite census and a named reduction. No RH claim. No L^* claim.

1 Recollection

Write $E_{ij} = \sqrt{v_i v_j}\, K_n^\mu(y_i, y_j)$ for the IIKS Gram at the ν-nodes (the Uvarov image of [1]). On FRAME-A $w=9$, $k=183$: $\lambda_{\max} = 0.999830$, maxdiag $= 0.9614$, assist $= 0.0399$, relative Gershgorin $g_A = 13.32$, cancellation 0.9970. Gershgorin of the absolute matrix is too crude [2]. The remaining question is whether the off-diagonal *sign pattern* is structural enough to force $\lambda_{\max} \le \mathrm{maxdiag}\cdot(1+o(1))$ from the source (F1-runs, separation, $\rho_{\mathrm{AP}} < 1/5$).

The candidate: order the ν-nodes by angle and hope $\mathrm{sign}(K_{ij}) = (-1)^{i-j}$ (checkerboard), equivalently a sign vector σ with $\sigma_i\sigma_j K_{ij} \le 0$ for almost all pairs. Then $D_\sigma E D_\sigma$ would be a Z-matrix, and M-matrix / Perron theory would bound $\lambda_{\max}$.

2 Dirichlet signs on the cosine grid

The Chebyshev-T Christoffel–Darboux kernel is [2]

$$K_n^T(\theta,\phi) = \frac{D_{n-1}(\theta-\phi) + D_{n-1}(\theta+\phi)}{2\pi}, \qquad D_m(\alpha) = \frac{\sin\big((m+\frac{1}{2})\alpha\big)}{\sin(\alpha/2)}.$$

Lemma 1 (Dirichlet sign)**.** *For $\alpha \notin 2\pi\mathbb{Z}$,*

$$\operatorname{sign}(D_m(\alpha)) = \operatorname{sign}\big(\sin((m+\tfrac{1}{2})\alpha)\big) \cdot \operatorname{sign}\big(\sin(\alpha/2)\big).$$

On the uniform cosine grid $\theta_j = \pi j/S$ the difference term is zonal: its sign depends only on which π-interval contains $(m+\frac{1}{2})|\theta_i - \theta_j|$. Machine identity on $S = 16, 12, 21$.

Lemma 2 (Dirichlet envelope)**.** $|D_n(\alpha)| \le \min\big(2n+1,\ 1/|\sin(\alpha/2)|\big)$ *for $\alpha \notin 2\pi\mathbb{Z}$. Equality of the pointwise ratio to* 1 *is attained on the tested mesh; the bound is classical.*

The reflected sum $D_{n-1}(\theta_i + \theta_j)$ flips about one eighth of the pair-signs on a small grid (agree 0.875 against the difference term alone). The zonal law is a theorem of the Chebyshev kernel, not of the μ-OP Gram.

3 The hoped bound is false in 2×2

Lemma 3 (Z-matrix bound false)**.** *For $0 < c < 1$ the matrix $\left(\begin{smallmatrix} 1 & -c \\ -c & 1 \end{smallmatrix}\right)$ is positive definite with negative off-diagonals and $\lambda_{\max} = 1 + c >$ maxdiag. A diagonal sign conjugation to a Z-matrix does* not *imply $\lambda_{\max} \le$ maxdiag.*

The 2×2 already kills the M-matrix/Perron slogan as a closing deduction, even granting perfect conjugacy. What remains is a *quantitative* remainder: if the positive off-diagonal mass after a *source-defined* conjugation is o(maxdiag), Weyl's inequality still needs a bound on $\lambda_{\max}$ of that remainder. Oracle conjugation by the true top eigenvector of E on FRAME-A leaves a positive-remainder row-sum $1.90 > 1$.

4 The sign map, measured

Proposition 1 (sign census)**.** *On FRAME-A $w = 9$ at $k = 183$, ordering ν-nodes by angle:*

- *checkerboard mass-weighted agreement* 0.504 *(coin);*
- *Chebyshev-CD mass-weighted agreement* 0.810*, pair-count* 0.721 *(small entries flip);*
- *nearest-neighbour negative mass-share* 0.468 *(not an M-matrix path);*
- *last twelve Chebyshev mass-agreements in* $[0.801, 0.829]$ *(robust in k, not a SATZ);*
- *at the r382 entry $k = \lfloor 2N/5 \rfloor = 73$ the nearest neighbours are* constructive *(negative mass-share* 0.039*).*

*Core-*42 *at the wall: Chebyshev mass-agreement in* $[0.810, 0.894]$*, checkerboard in* $[0.429, 0.512]$*, nearest-neighbour negative share in* $[0.349, 0.543]$*,* 42/42 *with $\lambda_{\max} < 1$. Builder fallback: core-*42 *only.*

The τ-deformation of r392 is small in the *formula* for γ, but not small enough to transfer the Dirichlet-zone SATZ to K^{μ}: eleven to nineteen percent of the off-diagonal *mass* disagrees, and the pair-count disagrees at one quarter to one third. The large couplings follow Chebyshev; the bipartition does not.

5 Adversaries

Lemma 4 (two-period in-phase)**.** *On the two-period mesh $S = 81$, $c = \frac{2}{3}$, at depth $k = 22$ every nearest-neighbour off-diagonal of E is positive (40/0) and $\lambda_{\max} = 1.0288 > 1$. A checkerboard conjugation along the chain is impossible: the short-lag couplings are in-phase. At $k = 40$ the Chebyshev (and μ-CD) sign pattern* is *exactly the anti-checkerboard, but too late:* maxdiag $= 1.317$ *already exceeds* 1*.*

Named kills. Scramble occupation seed 1 (the r388 $\mathrm{d}\Delta = 2.16 \cdot 10^6$ adversary) keeps Chebyshev mass-agreement 0.788 and diverges at $\lambda_{\max} = 9.17 \cdot 10^6$: it breaks the *envelope* (occupation / maxdiag), not the mass-weighted sign map. The mutant $E \mapsto |E|$ (signs dropped) yields $\lambda_{\max}(|E|) = 1.683 > 1$ on FRAME-A; the relative $g_A = 13.32$ is sign-blind and unchanged. Assist 0.0399 is entirely the signed remainder.

6 What remains

Proposition 2 (reduction)**.** *Sign-Schur as a rank-1 / checkerboard / M-matrix fact is REFUTED as a closing deduction. The Chebyshev-CD signs on the cosine grid are Dirichlet-zonal (SATZ); $K^\mu[\Xi]$ inherits those signs in the large couplings (census, not a SATZ) and not as a bipartition. No source-defined conjugation bound improves the r382 entry: the checkerboard positive-remainder majorant crosses* 1 *at $k = \lfloor 2N/5 \rfloor$, and the oracle bound stays above* 1 *through the wall. Every $k^* > 2N/5$ would have improved* `adaptive_band_from_entry` *mechanically; this round does not produce one. The Assist contraction remains the signed μ-OP Gram of r387/r389, now with a named negative: it is* not *a Schur bipartition of the ν-nodes. The hydra-legal remainder is the Dirichlet/Abel (Weyl) energy of that Gram — the r389 rest — not a sign vector. F_ε (bound on $\Delta^2 \log \tau$ under F_1) and W_ε are untouched. No m_4. No RH claim.*

7 Machine checks

Numbered lemmas: `verify_sign_schur.py` (same directory). Standalone: Dirichlet sign and envelope; Chebyshev-CD; 2×2 Z-matrix; bookkeeping; checkerboard rank-1; mesh step. Construction: FRAME-A Assist, Chebyshev/checkerboard mass-agreements, $|E|$ mutant, two-period in-phase and too-late anti-checkerboard, scramble envelope. Discovery census: `sign_schur_probe.py` (`--smoke` / full), core-42. Exit line: `SIGN SCHUR VERIFIED`.

Claim boundary

Finite identities for the Chebyshev–Dirichlet kernel on a cosine grid, a named 2×2 counterexample to the hoped M-matrix bound, and named two-period / scramble / unsigned breaks of a rank-1 sign conjugation. No statement about zeros of $\zeta(s)$, in either direction. No RH claim. No L^* claim.

References

[1] S. Hamann, *The deletion transform, reduced*, standalone note, August 2026, `rh/problem/deletion_transform.tex`.

[2] S. Hamann, *Coherence assist, reduced*, standalone note, August 2026, `rh/problem/coherence_assist.tex`.

[3] S. Hamann, *The Weyl energy of cancellation, reduced*, standalone note, August 2026, `rh/problem/weyl_energy.tex`.

[4] S. Hamann, *The Δ-deformation, reduced*, standalone note, August 2026, `rh/problem/delta_deformation.tex`.

[5] S. Hamann, *The pivot-band entry lemma, reduced*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.